Chapter 10. Mass-storage structure. Exercises

Original problems. State assumptions, units, and intermediate reasoning. Answers are in the separate solutions PDF.

OS10-E01. Disk scheduling

The head starts at cylinder 50, initially toward larger numbers. Pending requests are 82,170,43,140,24,16,190.

1. Give FCFS order.
2. Give SSTF order starting from 50.
3. Explain SSTF starvation.

Check your approach. Show the service sequence and the distance calculation under the stated model.

OS10-E02. RAID 5 and mirror groups

Compare four equal disks used as one RAID 5 array with the same disks used as two independent two-disk RAID 1 mirrors. Count both mirror groups' total capacity. No striping between mirror groups is assumed; ignore controller cache and scrubbing.

1. What is the RAID 5 data capacity?
2. What is the total mirror-group data capacity?
3. Compare one-disk fault recovery and state the conditions.

Check your approach. Keep array organization, capacity, and tolerated failure patterns distinct.

OS10-E03. A stable write

One second after a write reports success, power fails. On restart the final 512 bytes are missing.

1. What did success promise if the call only updated the page cache?
2. What can establish a stronger durability condition?
3. Is a mirror sufficient by itself?

Check your approach. Distinguish cache acceptance, device acknowledgment, and durable commit.

Chapter 10. Mass-storage structure. Exercises (continued)

OS10-E04 RPM, IOPS, and response

Teaching HDD:seek 5 ms,7200 RPM, uniform phase,64 KiB request,128 MiB/s transfer, queue wait 15 ms; one-at-a-time, no cache.

1. Find rotation, transfer, and service.
2. Find response and serial IOPS.
3. Find bandwidth at 1000 IOPS for 4 KiB and 64 KiB.

OS10-E05 Flash mappings and WA

4 KiB pages, four-page erase blocks. B0:P0…P3 contain L0…L3; B1:P4…P7 erased. Overwrite L1 and reclaim B0.

1. Show out-of-place update and post-GC mappings.
2. Compute WA without discard and with L2 discarded first.
3. What do discard and crash safety separately require?

OS10-E06 LOOK, SCAN, and wrap cost

Cylinders 0…199, head 53, all requests 98,183,37,122,14,124,65,67 arrive 0. Count physical wrap until the last request completes.

1. Give LOOK-down and SCAN-down paths/totals.
2. Does SSTF total 236 prove it is the same as SCAN?
3. Give upward C-SCAN/C-LOOK totals and starvation premises.

OS10-E07 Parity, write holes, and independent equations

RAID5 D0=3C, D1=A5, D2=5A, all hex. Change D1 to A4. Separately RAID6 uses GF256/modulus 11D, coefficients 1/2/4.

1. Find old/new parity and reconstruct missing D1.
2. New D1 is durable but parity C3 remains, then D2 fails. Show the wrong reconstruction.
3. Recover D1/D2 with the specified field example.
4. Prove two independent equations suffice. Does duplicating XOR suffice?

OS10-E08 Two-copy crash prefixes

Serialized 512-byte record:4-byte generation,504-byte payload,4-byte CRC. Copies begin v1; write/flush C1, then C2, then ACK.

Chapter 10. Mass-storage structure. Exercises (continued)

1. Which versions survive a crash during C1 or C2?
2. Give the concrete format and CRC/version selection rule.
3. Prove the invariant and specify repair order under another crash.
4. May unacknowledged work recover new? Is a CPU barrier persistence?

OS10-E09 Topology, interfaces, and swap references

Four disks A/B/C/D. 0+1 stripes AB/CD then mirrors; 1+0 mirrors AB/CD then stripes. A SAN LUN and swap S7/value 41 referenced by P1/P2 are separate examples.

1. AfterA thenC orB fail, which arrays remain readable?
2. Where doNAS/SAN resolve file offsets? Can two uncoordinated ordinary filesystems mount oneLUN for writes?
3. P1 reads S7 and privately writes 42 while P2 still references S7. May S7 be overwritten/reclaimed?
4. With 512-byte sectors, giveA=[1,101) MiB andB=[101,301) MiB boundaries.

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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https://github.com/xinhi6716/computer-science-self-study